NILENDU ADHIKARY

EDUCATION

Indian Institute of Technology (BHU), Varanasi *2023 - 2028*
Integrated Dual Degree in Computer Science and Engineering *8.41*

CHANSERPUR HIGH SCHOOL *2023*
WBCHSE *93.20%*

CHANSERPUR HIGH SCHOOL *2021*
WBBSE *96.20%*

SKILLS

- **Programming:** C/C++, Python, SQL, JavaScript, LaTeX
- **Frameworks & Libraries:** PyTorch, scikit-learn, Pandas, NumPy, Flutter, Django, Flask, ReactJS, Node.js
- **Tools & Databases:** Git, Streamlit, Unix Shell, VS Code, MySQL, MongoDB

PROJECTS

Kids Magazine Android App *Google Play Store*
- Developed a multilingual learning app to help children learn native scripts.
- Built a custom Hindi transliteration system and integrated Coqui TTS (Bengali) and gTTS (Hindi), including long-text handling support.
- **Tech Stack:** Flutter, Dart, Python, PyTorch, Flask, Firebase.

GraphFusion: Multimodal Hateful Meme Understanding using Graph Reasoning
- Developed a multimodal hateful meme classification framework integrating transformer-based text embeddings, CLIP visual features, and graph neural networks for robust meme understanding.
- Designed a hybrid fusion architecture with **GATv2-based graph reasoning, contradiction-aware fusion and dynamic modality gating** to capture complex cross-modal semantic interactions.
- **Accepted At ACM-ICMI 2026 :** Manuscript.
- **Tech Stack:** PyTorch, PyTorch Geometric, Transformers, CLIP, GATv2, Contrastive Learning, SpaCy.

Delta Hedging & Adversarial Stock Path Generation Simulator *May 2026 – Jun 2026*
- Built a quantitative finance simulator to generate **10K+ synthetic stock paths** and evaluate option hedging strategies under normal, stressed, and adversarial market conditions.
- Implemented **3 stock path generation techniques** and developed a **dynamic delta hedging engine** with 100+ rebalancing intervals, transaction costs, portfolio valuation, and P&L computation.
- **Tech Stack:** Python, NumPy, PyTorch, Monte Carlo Simulation, Black–Scholes Model, Delta Hedging

PUBLICATIONS

JustGen@LT-EDI 2026: Controlled Gender Inclusive and Bias-Aware Language Generation using LLMs.
May 2026
- Developed a multilingual Retrieval-Augmented Generation (RAG) framework for transforming gender-biased text into inclusive alternatives using FAISS, Mistral-7B, and LLaMA.
- Designed a hybrid pipeline combining lexical bias correction, semantic retrieval, edit-distance constrained generation, and self-verification to preserve meaning while improving inclusivity.
- Accepted at ACL Workshop LT-EDI.
- Tech Stack: Python, FAISS, Mistral-7B, LLaMA, Sentence Transformers, RAG, Prompt Engineering.

ACHIEVEMENTS

- **Academic Achievements:** AIR 252 in WBJEE 2023, AIR 62 in NEST 2023.
- **Competitive Programming:** Achieved peak ratings of 1715 on Codeforces and solved 400+ algorithmic problems on LeetCode.
- Secured **Global Rank 226** in **Codeforces Div. 3 Round 1084**.
- Achieved a **Perfect Performance Index (PI) of 10.0,** securing a branch change from **Mechanical Engineering** to **CSE** based on first-Semester academic performance.

POSITIONS OF RESPONSIBILITY

Codefest *Sep 2024– Feb 2025*
- Served as an Executive in the Marketing Team, contributing to sponsorship outreach and fundraising initiatives.

EXTRA CURRICULARS

Conferences & Volunteering
- Attended and volunteered at **FIRE 2025 (Forum for Information Retrieval Evaluation),** contributing to conference operations, attendee coordination, and event management.

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